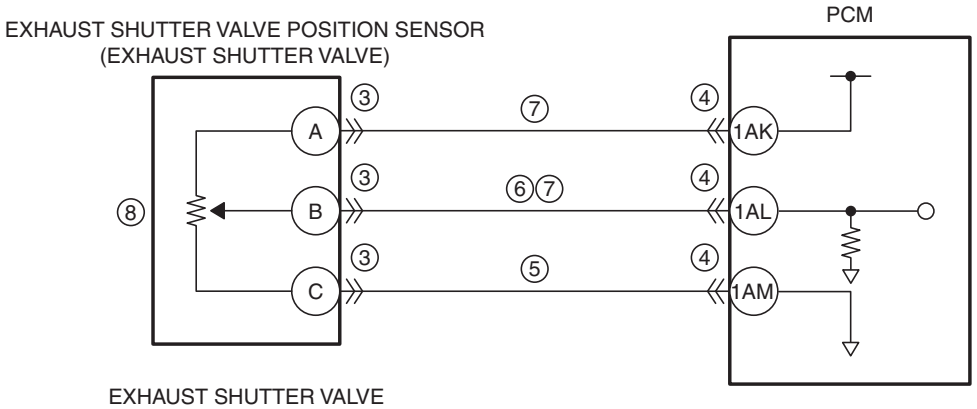
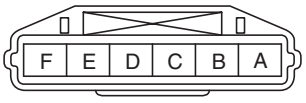
DTC P048E:00 [SKYACTIV-D 1.5]

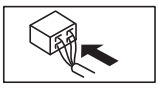
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DTC P048E:00	Exhaust shutter valve position sensor circuit high input
DETECTION CONDITION	• If the input voltage at the PCM terminal 1AL is **more than 4.77 V** for **5 s**, the PCM determines that the exhaust shutter valve position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after ignition was switched on. — PCM internal processor malfunction DTC P0606:00 not detected. **Diagnostic support note** • This is a continuous monitor (EGR system). • The check engine light illuminates if the PCM detects the above malfunction condition during the first drive cycle. • FREEZE FRAME DATA/Snapshot data is available. • DTC is stored in the PCM memory.
FAIL-SAFE FUNCTION	• PCM restricts engine torque. • Inhibits the EGR control.
POSSIBLE CAUSE	• Exhaust shutter valve connector or terminals malfunction • PCM connector or terminals malfunction • Open circuit in wiring harness between exhaust shutter valve terminal C and PCM terminal 1AM • Short to power supply in wiring harness between exhaust shutter valve terminal B and PCM terminal 1AL • Exhaust shutter valve position sensor power supply circuit and signal circuit are shorted to each other • Exhaust shutter valve position sensor (built-into exhaust shutter valve) malfunction • PCM malfunction

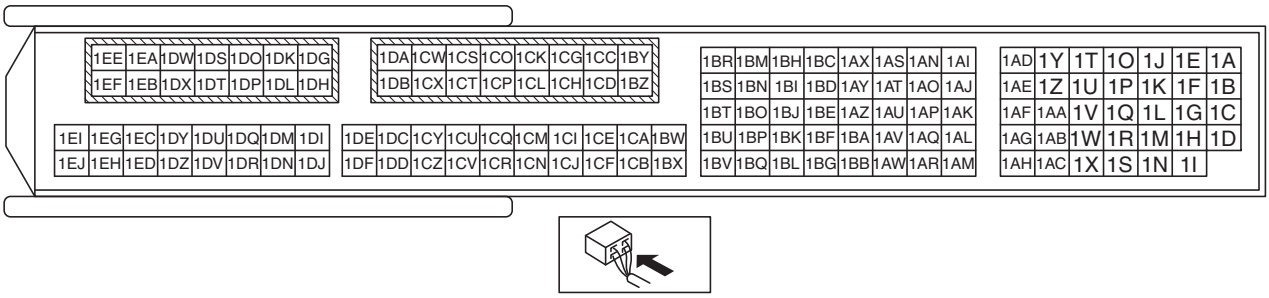


EXHAUST SHUTTER VALVE WIRING HARNESS-SIDE CONNECTOR





PCM WIRING HARNESS-SIDE CONNECTOR



Diagnostic Procedure

STEP	INSPECTION		ACTION
1	**RECORD VEHICLE STATUS WHEN DTC WAS DETECTED TO UTILIZE WITH REPEATABILITY VERIFICATION** **Note** • Recording can be facilitated using the screen capture of the PC function. • Record the freeze frame data/snap shot data.	—	Go to the next step.
2	**VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related service information availability. • Is any related service information available?	Yes	Perform repair or diagnosis according to the available service information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
3	**INSPECT EXHAUST SHUTTER VALVE CONNECTOR CONDITION** • Switch the ignition off. • Disconnect the exhaust shutter valve connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
4	**INSPECT PCM CONNECTOR CONDITION** • Disconnect the PCM connector. • Inspect for poor connection (such as damaged/ pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 9.
		No	Go to the next step.
5	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR GROUND CIRCUIT FOR OPEN CIRCUIT** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Inspect for continuity between exhaust shutter valve terminal C (wiring harness-side) and PCM terminal 1AM (wiring harness-side). • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between exhaust shutter valve terminal C and PCM terminal 1AM. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 9.
6	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR SIGNAL CIRCUIT FOR SHORT TO POWER SUPPLY** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Switch the ignition ON (engine off). **Note** • Another DTC may be stored by the PCM detecting an open circuit. • Measure the voltage at the exhaust shutter valve terminal B (wiring harness-side). • Is the voltage **0 V**?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between exhaust shutter valve terminal B and PCM terminal 1AL. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 9.

STEP	INSPECTION		ACTION
7	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR POWER SUPPLY CIRCUIT AND SIGNAL CIRCUIT FOR SHORT TO EACH OTHER** • Verify that the exhaust shutter valve and PCM connectors are disconnected. • Switch the ignition off. • Inspect for continuity between exhaust shutter valve terminals A and B (wiring harness-side). • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Exhaust shutter valve terminal A—PCM terminal 1AK • Exhaust shutter valve terminal B—PCM terminal 1AL **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to each other. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to each other. Go to Step 9.
		No	Go to the next step.
8	**INSPECT EXHAUST SHUTTER VALVE POSITION SENSOR** • Reconnect all disconnected connectors. • Inspect the exhaust shutter valve position sensor. (See EXHAUST SHUTTER VALVE POSITION SENSOR INSPECTION [SKYACTIV-D 1.5].) • Is there any malfunction?	Yes	Inspect the exhaust shutter valve, then go to next step. (See EXHAUST SHUTTER VALVE INSPECTION [SKYACTIV-D 1.5].)
		No	Go to the next step.
9	**VERIFY DTC TROUBLESHOOTING COMPLETED** • Always reconnect all disconnected connectors. • Clear the DTC from the PCM memory using the M-MDS. (See CLEARING DTC [SKYACTIV-D 1.5].) • Perform the KOEO or KOER self test. (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) • Is the same DTC present?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the PCM. (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Go to the next step.
		No	Go to the next step.
10	**VERIFY AFTER REPAIR PROCEDURE** • Perform the “AFTER REPAIR PROCEDURE”. (See AFTER REPAIR PROCEDURE [SKYACTIV-D 1.5].) • Are any DTCs present?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [SKYACTIV-D 1.5].)
		No	DTC troubleshooting completed.